Dietary Linkage to Microbiome and Their Influence on Colorectal Cancer

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Abstract

Diet is known to influence the gut microbiome, but whether diet-associated microbial taxa contribute to the tumour microenvironment or to patient outcomes in colorectal cancer (CRC) remains unclear. This report integrated whole-genome sequencing (WGS) tumour-associated microbial profiles from The Cancer Genome Atlas (TCGA) with food-derived metagenomes from the curatedFoodMetagenomicData (cFMD) resource to evaluate potential diet–microbiome–survival relationships. Three research questions were examined: (1) whether survival in CRC is associated with enrichment of diet-linked microbial taxa, (2) whether survival-associated taxa overlap with specific food categories, and (3) whether tumour microbial profiles can predict survival outcomes.

Differential abundance analysis of 138 tumour samples (25 survivors, 113 deceased) identified taxa with apparent shifts between survivor and deceased groups. However, many of the taxa driving these differences were contamination-prone genera, and several showed effect directions that were biologically implausible. A UMAP embedding of shared tumour–food species showed no clustering by survival status, and a Fisher's exact test indicated only nominal statistical significance without ecological coherence. Cross-referencing survival-associated taxa with cFMD revealed extensive overlap between survivor-enriched taxa and dairy-associated microbes—particularly *Lactococcus lactis*—but this reflected ecological ubiquity in fermented dairy rather than verified dietary transfer to tumour tissue. Deceased-enriched taxa showed no meaningful food source associations.

Machine-learning models (Random Forest and CatBoost) trained on species-level tumour microbiome features exhibited low predictive power (AUC < 0.60) and severe overfitting, while tumour stage showed no detectable association with microbial composition. Across all analyses, species-level tumour microbiome profiles contained no reliable dietary signal and no clinically meaningful information related to survival.

Overall, this study finds no evidence that diet-linked microbial taxa influence survival in colorectal cancer within the TCGA cohort. The results highlight the challenges of analysing low-biomass tumour microbiomes, the importance of contamination-aware sequencing, and the need for strain-level functional analyses and validated dietary metadata in future research.

Abstrakt(Deutsch)

Die Ernährung beeinflusst die Zusammensetzung des Darmmikrobioms, doch ist bislang unklar, ob ernährungsassoziierte Mikroorganismen das Tumormikromilieu oder das Überleben von Patientinnen und Patienten mit kolorektalem Karzinom (CRC) beeinflussen. In dieser Arbeit wurden mikrobielle Profile aus Whole-Genome-Sequencing-(WGS)-Daten von Tumorgewebe aus The Cancer Genome Atlas (TCGA) mit lebensmittelassoziierten Metagenomen aus der curatedFoodMetagenomicData-(cFMD)-Datenbank integriert, um potenzielle Zusammenhänge zwischen Ernährung, Mikrobiom und Überleben zu untersuchen. Drei Forschungsfragen wurden adressiert: (1) ob das Überleben mit einer Anreicherung ernährungsassoziierter Mikroben im Tumor verbunden ist, (2) ob überlebensassoziierte Taxa bestimmten Lebensmittelkategorien zuzuordnen sind und (3) ob mikrobielle Tumorprofile eine Vorhersage des Überlebens ermöglichen.

Die differentiellen Abundanzanalysen von 138 Tumorproben (25 Überlebende, 113 Verstorbene) zeigten scheinbare Unterschiede zwischen den Gruppen. Viele der beteiligten Taxa waren jedoch für Kontamination anfällig oder zeigten biologisch nicht plausible Effektrichtungen. Eine UMAP-Einbettung der zwischen TCGA und cFMD geteilten Arten zeigte keine Clusterbildung in Bezug auf das Überleben, und ein Fisher's Exact Test ergab zwar eine nominale Signifikanz, jedoch ohne ökologische oder biologische Kohärenz. Der Abgleich der überlebensassoziierten Taxa mit der cFMD-Datenbank ergab eine deutliche Überschneidung mit milchfermentationsassoziierten Mikroben—insbesondere *Lactococcus lactis*—was jedoch eher deren ökologische Allgegenwärtigkeit widerspiegelt als eine tatsächliche Übertragung über die Ernährung. Für verstorbenen-assoziierte Taxa fanden sich keine konsistenten lebensmittelbezogenen Muster.

Maschinelle Lernmodelle (Random Forest und CatBoost), die auf artenbasierten Tumormikrobiomprofilen trainiert wurden, zeigten eine geringe Vorhersagekraft (AUC $< 0{,}60$) und starke Überanpassung. Ebenso lieferte die Tumorstadienanalyse keinen Hinweis auf eine mikrobielle Strukturierung. Insgesamt enthielten die verfügbaren Tumormikrobiomdaten weder robuste Ernährungssignaturen noch klinisch relevante Informationen zum Überleben.

Diese Arbeit zeigt, dass in der TCGA-Kohorte keine Hinweise darauf vorliegen, dass ernährungsassoziierte Mikroben das Überleben bei CRC beeinflussen. Die Ergebnisse verdeutlichen die Herausforderungen bei der Analyse mikrobiell armer Tumorgewebe, die Notwendigkeit kontaminationsbewusster Sequenzierungsstrategien sowie den Bedarf an zukünftigen Studien mit Stamm- und Funktionsauflösung und validierten Ernährungsdaten.

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1 Introduction

Colorectal cancer (CRC) is one of the most prevalent malignancies worldwide and remains a major contributor to global cancer morbidity and mortality. According to recent epidemiological assessments, CRC accounts for approximately 10% of all newly diagnosed cancers, making it the third most commonly diagnosed cancer type globally [1]. Despite significant progress in screening, surgical techniques, and adjuvant therapies, CRC continues to exhibit substantial heterogeneity in disease behavior, prognosis, and therapeutic response.

The Cancer Genome Atlas (TCGA) has provided a comprehensive molecular characterization of CRC, revealing recurrent alterations in *APC*, *TP53*, *SMAD4*, *KRAS*, and *PIK3CA*, and demonstrating that colon and rectal cancers share a largely convergent genomic landscape [2]. These genomic insights have advanced molecular subtype classification and improved our understanding of tumorigenic pathways. However, genomic features alone are insufficient to explain inter-individual variation in tumor progression, immune infiltration, and clinical outcome, suggesting an important role for non-genetic modulators.

Recent studies have highlighted the strong influence of the gut microbiome—comprising bacteria, archaea, viruses, and fungi—on host physiology and disease. The gut microbiota regulates metabolic functions, epithelial integrity, and immune homeostasis, while dysbiosis can promote carcinogenesis through mechanisms including inflammation, genotoxic metabolite production, and immune evasion [3]. Several microbial species have been implicated in CRC pathogenesis, such as *Fusobacterium nucleatum*, which promotes tumor progression by modulating E-cadherin–-catenin signaling and impairing T cell–mediated immunity, or enterotoxigenic *Bacteroides fragilis*, which induces DNA damage and drives pro-tumorigenic inflammation. Conversely, commensal taxa associated with anti-inflammatory activity—such as *Bifidobacterium* and certain *Lactobacillus* species—have been linked to beneficial immunomodulatory effects, although cancer-type–specific evidence remains emerging.

Large-scale, multi-omics studies have recently expanded the understanding of CRC-associated microbial communities by characterizing the intratumoral microbiome. Integrative analyses combining metagenomic, transcriptomic, and immune-profiling data have shown that the tumor microenvironment in CRC harbors a distinct and biologically coherent microbial signature [4]. This signature is reproducible across independent cohorts and includes taxa such as *Ruminococcus bromii*, whose enrichment has been associated with favorable immune infiltration and improved clinical outcome. Moreover, recent TCGA-wide reanalyses have clarified that, when appropriate contamination controls are applied, the intratumoral microbiome of CRC remains one of the most robust and biologically interpretable across all cancer types [5], [6]. These findings reinforce the hypothesis that specific microbial taxa influence tumor behavior and may serve as biomarkers for prognosis or therapeutic stratification.

Diet represents one of the strongest and most modifiable determinants of gut microbial composition. Long-term dietary intake shapes the availability of substrates for microbial fermentation, modulating metabolic outputs such as short-chain fatty acids, bile acid derivatives, and reactive metabolites [3]. Diets enriched in fiber promote butyrate-producing bacteria that enhance epithelial barrier integrity and regulate immune responses, whereas high-fat or processed diets foster pro-inflammatory microbial communities associated with increased CRC risk. Fermented foods—such as cultured dairy products, kimchi, sauerkraut, and fermented

meats—introduce live microbial communities dominated by lactic acid bacteria including *Lactobacillus*, *Leuconostoc*, and *Weissella*, which may transiently colonize or interact with the host intestinal mucosa. These fermentation-derived taxa have been proposed to influence immune function and inflammatory tone, suggesting a plausible interface between dietary microbes and tumor-associated microbial patterns.

The microbiomes of fermented and non-fermented foods have recently been systematically characterized through curated metagenomic resources such as the curatedFoodMetagenomicData (cFMD) project, which provides taxonomic profiles for thousands of food samples obtained via standardized shotgun metagenomic analysis [7]. These datasets offer a unique opportunity to explore whether food-associated microbes overlap with those detected in colorectal tumors and normal tissues. However, such an investigation requires rigorous harmonization of taxonomies, careful correction for compositional effects, and robust methods to align data generated from distinct biological matrices.

Compositional data analysis techniques are essential when analyzing sequencing-derived microbial profiles, as relative abundances are constrained by a constant-sum effect that induces spurious correlations [8]. Log-ratio transformations—particularly the centered log-ratio (CLR) transformation—provide a principled solution by projecting compositional profiles into an unconstrained Euclidean space suitable for downstream multivariate techniques [9], [10], [11]. Such transformations enable valid application of linear ordination methods, such as principal component analysis (PCA), and are now widely recommended for comparative microbiome studies.

Despite substantial progress in characterizing CRC-associated microbiomes, the relationship between tumor-resident microbial species and those found in the food microbiome remains poorly understood. Few studies have directly compared these domains, and none have systematically integrated species-level tumor microbiome data with a comprehensive food microbial reference. Questions concerning whether food-derived taxa appear in colorectal tumor tissue, whether fermentation-associated microbes cluster closer to normal or tumor microbiomes, and whether shared taxa exhibit differential enrichment in survivors compared to deceased patients remain largely unanswered.

The present study addresses this gap by integrating species-level microbial profiles from TCGA colorectal cancer tissues with food-associated metagenomic profiles from the cFMD resource. After taxonomic canonicalization, CLR transformation, and feature harmonization, both datasets were embedded into a shared multivariate space to evaluate cross-domain similarity. Statistical tests were applied to assess tumor–normal differences and survival-associated enrichment. In addition, machine learning models were trained to identify microbial taxa predictive of survival outcomes, and their intersection with food-derived microbes was examined. Through these analyses, the study aims to: (i) identify species shared between food microbiomes and CRC tissues; (ii) evaluate whether shared taxa exhibit differential enrichment between surviving and deceased CRC patients; and (iii) characterize the ecological proximity of fermented-food microbiomes to tumor- and normal-associated microbial communities.

By integrating dietary and tumor microbiome datasets within a unified analytical framework, this work contributes to an emerging research direction exploring potential

diet–microbiome–cancer interactions. The approach highlights how food-associated microbial reservoirs may relate to tumor-resident communities and offers a reproducible platform for future investigations at the interface of nutrition, microbiology, and cancer biology.

2 Research Questions and Objective

2.1 Research Questions

This project investigates whether microbial taxa originating from food-associated environments are detectable within colorectal tumour microbiomes, and whether these diet-linked taxa show any association with patient outcome. The analysis is structured around three focused research questions:

1. **RQ1: Is survival in colorectal cancer associated with enrichment of diet-linked microbial taxa?**
   This question tests whether species shared between food metagenomes (cFMD) and CRC tumour microbiomes (TCGA) differ in abundance between survivors and deceased patients.

2. **RQ2: Are microbial taxa enriched in survivors also enriched in specific food categories?**
   This evaluates whether taxa associated with favourable survival outcomes overlap with microbial signatures characteristic of particular food categories (e.g. fermented, non-fermented, animal-derived, plant-based).

3. **RQ3: Can diet-associated microbial taxa predict survival in colorectal cancer?**
   This question assesses whether microbial features—especially those shared between food and tumour datasets—carry predictive value for survival in supervised machine learning models.

2.2 Objectives

The overarching objective of this project is to integrate, harmonise, and jointly analyse tumour-associated microbiome profiles from The Cancer Genome Atlas (TCGA) with food-associated metagenomic data from the curatedFoodMetagenomicData (cFMD) resource, in order to determine whether diet-linked microbial taxa contribute to survival-associated signatures in colorectal cancer (CRC).

To address the research questions above, the project pursues the following methodological objectives:

1. **Integrate TCGA microbial profiles with clinical metadata**
   Merge species-level tumor microbiome profiles (S4) with TCGA sample metadata (S12) and curated clinical annotations (survival status, demographics), followed by CLR transformation and batch correction to obtain a unified analytical dataset.

2. **Preprocess and harmonise food microbiomes**
   Process cFMD species-level annotations, canonicalise taxonomic labels, and construct a cleaned species-by-sample abundance matrix aligned with TCGA taxonomy.

3. **Identify shared microbial taxa across tumor and food datasets**
   Derive a canonical set of species present in both datasets, representing the intersection of tumor- and food-associated taxa for downstream comparative analyses.

4. **Quantify differential abundance related to survival**

Compute mean relative abundances for survivors and deceased patients, derive species-level difference metrics (Δ_t), and assess whether shared food–tumor taxa show systematic enrichment patterns.

5. **Embed tumor and food samples into a joint feature space**
   Project CLR-transformed tumor and food microbiome profiles into a shared PCA or UMAP embedding, compute cross-domain similarities, and evaluate whether food categories occupy regions proximal to tumor microbial structure.

6. **Evaluate the predictive value of diet-linked microbial taxa**
   Train supervised learning models (Random Forest, CatBoost) on microbial features to test whether tumor-associated or food-linked taxa contribute predictive signal for CRC survival.

3 Datasets and Data Preparation

3.1 TCGA Colorectal Cancer Microbiome Dataset

The colorectal cancer microbiome data were obtained from The Cancer Genome Atlas (TCGA) colon adenocarcinoma (COAD) and rectal adenocarcinoma (READ) cohorts [2]. Raw whole-genome sequencing (WGS) reads from these projects were reprocessed in large-scale metagenomic analyses [5], [6], which applied stringent contamination filtering, taxonomic profiling and uniform normalization across all TCGA samples. The resulting microbial abundance tables and associated metadata were distributed as the `S4` and `S12` files, respectively, both of which were used in this report.

3.1.1 Microbial Abundance Matrix

The `S4` file contains processed species-level microbial abundances, expressed as counts per million (CPM) for each sample. All species were taxonomically annotated using the NCBI taxonomy framework and harmonized across samples. Zero counts were retained, and a small pseudocount was added before CLR transformation to avoid undefined log operations. This table served as the quantitative foundation for ordination, embedding, and prevalence analyses.

3.1.2 Clinical and Sample Metadata

The corresponding `S12` table contains the technical and clinical metadata linked to each microbiome sample in the `S4` matrix. Each record represents a unique sequencing aliquot, annotated with multiple identifiers and experimental descriptors used for alignment and filtering.

Key fields include:

- `project_id` — TCGA project identifier specifying cancer type (e.g., `TCGA-COAD` for colon adenocarcinoma, `TCGA-READ` for rectal adenocarcinoma).
- `sample_type` — tissue classification (e.g., `primary_tumor`, `solid_tissue_normal`, `blood_derived_normal`).
- `aliquot_submitter_id` — unique sequencing aliquot code used to join clinical and microbial tables.
- `sequencing_center` — sequencing facility (e.g., `BI`, `WUGSC`, `Harvard_GCC_02`), included to assess potential batch effects.
- `rd_len`, `tot_aln`, `grch38_unmapped` — technical quality metrics describing read length, total aligned reads, and unmapped proportions.

For this report, only colorectal samples (`TCGA-COAD`, `TCGA-READ`) were retained. Blood-derived and control samples were excluded to minimize contamination artifacts and focus on tumor- and tissue-associated microbiomes. The metadata were used to annotate the microbiome matrix with relevant sample characteristics, enable tissue-type stratification (tumor vs. normal), and control for sequencing-center variability during downstream analyses.

3.1.3 Integration and Data Alignment

To merge both datasets, the `HopkinsID` variable was used as a shared unique identifier, ensuring one-to-one alignment between microbial and clinical records. This integration enabled stratification of the microbial profiles by tissue type (tumor vs. normal), sequencing center,

and patient characteristics, providing the basis for downstream statistical and comparative analyses.

3.1.4 Curated TCGA Clinical Dataset (ACG Team Repository)

To complement the standard TCGA metadata (`S12`), a harmonized and preprocessed clinical dataset was retrieved from the Applied Computational Genomics (ACG) Team's open-access repository (`https://github.com/acg-team/shared_files`). This curated table provides standardized patient-level and tumor microenvironment annotations for The Cancer Genome Atlas (TCGA) projects, derived directly from the Genomic Data Commons (GDC) and aligned to official TCGA identifiers.

Key variables include:

- `Tumor_Sample_Barcode` — the unique TCGA sample identifier used to merge clinical attributes with microbiome aliquots by matching the first 16 barcode characters.
- `sample_type` — indicates whether the tissue was a `primary_tumor`, `solid_tissue_normal`, or other category.
- `age` and `sex` — basic demographic information for each patient.
- `alive` — binary survival status (1 = alive, 0 = deceased) derived from vital status and overall survival metrics.
- `immune`, `stromal`, `estimate`, and `purity` — quantitative scores reflecting immune cell infiltration, stromal content, and tumor purity based on expression-derived estimates.
- `number_of_TIL_patches`, `til_percentage`, and `macrophages` — histopathology-derived immune metrics quantifying tumor-infiltrating lymphocytes (TILs) and macrophage density.
- `global_pattern` and `number_of_clusters` — categorical descriptors summarizing overall immune–stromal spatial organization within the tumor microenvironment.

All records were pre-filtered to include only colorectal cancer samples (`TCGA-COAD`, `TCGA-READ`). The integration with the microbiome abundance matrix (`S4`) was performed by aligning the `Tumor_Sample_Barcode` to the first 16 characters of the `aliquot_submitter_id` in the microbiome dataset, ensuring one-to-one correspondence between microbial and clinical profiles.

This merged dataset provided a comprehensive clinical–microbiome framework linking microbial composition to patient demographics, tumor type, immune infiltration, and survival outcome. Such integration was crucial for downstream analyses comparing microbial profiles between survivors and non-survivors and for assessing associations between tumor microbiome diversity and immune microenvironment characteristics.

3.2 Food Microbiome Dataset (cFMD)

The food-associated reference microbiome data were obtained from the *curatedFoodMetagenomicData (cFMD)* project curated by the Segata Lab [7]. This publicly available resource compiles uniformly processed shotgun metagenomic data from diverse food matrices, including fermented and non-fermented products, beverages, meats, grains, and plant-based items. All samples were processed through the same taxonomic and functional profiling pipelines, ensuring methodological compatibility with the TCGA metagenomic datasets and enabling direct cross-domain comparisons between food and tumor-associated microbiomes.

3.2.1 Metagenome-Assembled Genomes (MAGs)

The `cFMD_MAGs.tsv` file provides genome-level annotations for assembled microbial genomes recovered from food samples. Each row represents a single metagenome-assembled genome (MAG) characterized by both taxonomic and quality metrics. Important columns include:

- `genus` and `species` — NCBI-aligned taxonomic identifiers at genus and species resolution.
- `genome_size` — estimated genome size in base pairs.
- `n_contigs` and `N50` — assembly statistics describing fragmentation and contiguity.
- `completeness` and `contamination` — CheckM-derived estimates assessing genome quality and purity.

These annotations were used to filter and retain only high-quality genomes (`completeness` = 100%)) for downstream species-level comparisons and taxonomic alignment with TCGA microbial taxa.

3.2.2 Sample Metadata

The `cFMD_metadata.tsv` file provides contextual metadata for each food sample, describing its biological origin and processing characteristics. Key fields include:

- `sample_id` — unique sample identifier linking metadata to the corresponding MAG entries.
- `type` — broad food category (e.g., `meat`, `fish`, `fruits_and_vegetables`, `dairy`).
- `fermented/non-fermented` — binary classification indicating whether the product underwent microbial fermentation (`F`) or not (`NF`).
- `origin` — descriptive field capturing the production or environmental source of the sample.
- `curator` and `filtered` — metadata curation status, ensuring traceability and standardization within the cFMD framework.

The cFMD dataset thus provides both genomic and contextual descriptors of food-associated microbial communities, allowing biological inference at multiple levels of resolution. All species names were canonicalized to ensure taxonomic consistency with the TCGA dataset, and overlapping taxa were extracted to construct a shared feature space. Samples were categorized as *fermented* (F) or *non-fermented* (NF) for comparative analysis of potential fermentation-linked taxa observed in colorectal cancer tissue.

3.2.3 Data Harmonization and Normalization

To ensure comparability across datasets, all microbial taxa were *canonicalized* using a custom normalization function that converts species names to lowercase ASCII characters and removes substrain or descriptive suffixes (e.g., "subsp.", "strain"). This step minimizes redundancy caused by inconsistent naming conventions across metagenomic databases and facilitates consistent taxonomic alignment between datasets.

Raw microbial read counts were first converted to relative abundances by dividing each count by the total number of reads per sample, thereby standardizing for sequencing depth. Subsequently, the data were transformed using the *centered log-ratio (CLR)* transformation, which is recommended for compositional microbiome analysis [8], [9]. The CLR transformation projects relative abundances into a log-ratio space, allowing multivariate methods such as principal component analysis (PCA) to operate under valid assumptions of Euclidean geometry. This approach mitigates spurious correlations inherent to compositional data and

enables robust comparison of microbial community structures across datasets [10], [11].

3.3 Integration and Biological Context

After normalization and alignment, both datasets were projected into a shared feature space for principal component analysis (PCA) and clustering. The *fermented/non-fermented* attribute from the cFMD metadata was included to test whether microbial communities from fermented foods cluster closer to normal or tumor tissue samples in the embedded space. These embeddings enabled quantitative assessment of whether food-associated microbial communities occupy regions of the PCA space proximal to tumor or normal tissue samples.

All preprocessing and harmonization were implemented in Python using `pandas`, `numpy`, and `scikit-learn`. Reproducible code notebooks are available in the project repository.

4 Methodology

This chapter describes the analytical workflow used to link tumor-associated microbiomes in colorectal cancer (CRC) to food-associated microbiomes, and to explore potential associations between microbial profiles and patient survival. The underlying datasets and their structure are introduced in Chapter 3; here, the focus is on data handling, normalisation, statistical inference, and machine learning.

4.1 Overview of Analysis Pipeline

The analysis pipeline consisted of the following main stages:

1. Preprocessing of TCGA microbiome abundance data and associated clinical metadata.
2. Preprocessing and taxonomic harmonisation of the cFMD food microbiome dataset.
3. Construction of a shared species feature space between tumor and food microbiomes.
4. Cross-domain embedding of food and tumor microbiomes using principal component analysis (PCA), proximity indices, and cosine similarity.
5. Statistical comparisons of microbiome composition, including tumor–normal differential abundance testing, survival-based enrichment analysis, and Fisher exact testing of shared TCGA–cFMD taxa.
6. Supervised machine learning (Random Forest and CatBoost) to predict survival from tumor microbiome profiles and to identify taxa most strongly associated with outcome.

All analyses were implemented in Python (version 3.11) using `pandas` for data handling, `numpy` for numerical operations, `scikit-learn` for statistical learning and PCA, and `matplotlib`/`seaborn` for visualisation.

4.2 Microbiome Preprocessing

4.2.1 Taxonomic Canonicalisation

Species names from both TCGA (`S4`) and cFMD were harmonised using a custom canonicalisation function. This function converted names to lowercase ASCII, replaced underscores with spaces, removed substrain and descriptive suffixes (e.g. *subsp.*, *strain*, *biovar*), stripped punctuation, and collapsed multiple whitespaces to a single space. Columns with identical canonical names were aggregated by summing their abundances. This procedure reduces redundancy caused by inconsistent naming conventions and facilitates reliable taxon matching across datasets, consistent with broader recommendations to ensure standardised and reproducible taxonomic handling in large-scale metagenomic studies [12].

4.2.2 Relative Abundance and CLR Transformation

For all count-like microbiome tables, raw abundances were converted to relative abundances at the sample level by dividing each taxon's count by the total counts per sample. This step standardised for differences in sequencing depth. Subsequently, relative abundances were transformed using the centred log-ratio (CLR) transformation:

$$\mathrm{clr}(x_i) = \log\left(\frac{x_i}{g(x)}\right), \qquad g(x) = \left(\prod_{j=1}^{D} x_j\right)^{1/D},$$

where x_i is the relative abundance of taxon i in a given sample, and $g(x)$ is the geometric mean of all D taxa in that sample. Small pseudocounts were used to avoid taking the logarithm of zero.

Treating microbiome data as compositional and applying log-ratio transformations is now considered best practice to avoid spurious correlations and to place the data in a valid Euclidean space for downstream multivariate analysis [8], [9]. CLR-transformed matrices were therefore used as the basis for PCA, cosine similarity, and most statistical modelling steps.

4.2.3 Prevalence Filtering and Feature Selection

Within each dataset (TCGA, cFMD), taxa with zero counts across all samples were removed. For the shared analysis, only species present in at least a minimal fraction of samples (prevalence threshold) in either dataset were retained, based on per-taxon prevalence:

$$\mathrm{prev}(t) = \frac{1}{N}\sum_{n=1}^{N} I\{x_{nt} > 0\},$$

where x_{nt} is the count for taxon t in sample n, N is the number of samples, and $I\{\cdot\}$ is the indicator function. This filtering removed extremely rare taxa that are unlikely to yield robust signals and reduced the dimensionality of the CLR feature space.

4.3 TCGA–Clinical Integration and Batch Handling

4.3.1 Linking Microbiome, Metadata and Clinical Outcomes

Microbial abundance data from `S4` were linked to the TCGA sample and project metadata in `S12` via the `HopkinsID` identifier. Only samples from the colon adenocarcinoma (COAD) and rectal adenocarcinoma (READ) projects were retained, and blood-derived or control samples were excluded to focus on tissue-associated microbiomes. In a second step, a curated TCGA clinical dataset from the ACG Team repository [13] was merged by matching the first 16 characters of the `Tumor_Sample_Barcode` to the corresponding aliquot identifiers in `S12`.

This yielded a unified table containing, for each primary tumour sample: CLR-normalised microbial abundances; sample-type labels (e.g. primary tumour, solid tissue normal); project identifiers; demographic variables (age, sex); immune and stromal scores; and binary survival status (alive vs. deceased) derived from overall survival information.

4.3.2 Batch Effect Correction Across Sequencing Centres

Because TCGA samples were generated at multiple sequencing centres, the presence of systematic technical variation was first assessed by inspecting PCA embeddings of the CLR-transformed microbial abundance matrix, coloured by `sequencing_center`. Distinct centre-specific clustering indicated non-biological structure consistent with batch effects, a well-recognised issue in microbiome sequencing workflows. Technical artefacts arising from reagent

lots, extraction kits, and sequencing runs can introduce variation that surpasses biological signal, making batch correction essential for reliable downstream inference [14].

To mitigate these centre-specific artefacts, batch correction was performed using the ComBat empirical Bayes framework [15]. ComBat models batch effects as additive and multiplicative shifts and estimates centre-specific parameters by pooling information across features, thereby stabilising variance in high-dimensional microbial abundance matrices. ComBat has been widely adopted for sequencing-derived datasets and has repeatedly been shown to outperform simple linear regression approaches for correcting multi-centre technical variability.

In this study, ComBat was applied to the CLR-transformed TCGA abundance matrix using `sequencing_center` as the batch variable. The resulting batch-adjusted microbial profiles were subsequently used for PCA, statistical analyses, and machine learning modelling, ensuring that downstream results reflected biological rather than technical variation.

4.4 Food Microbiome Preprocessing and Alignment

Food-associated microbiome data from curatedFoodMetagenomicData (cFMD) were processed analogously. MAG-level annotations were used to identify high-quality genomes based on completeness and contamination, and species-level taxonomic labels were canonicalised using the same function as for TCGA. Sample metadata provided food type and fermentation status (fermented, non-fermented), which were used to define categories such as *fermented dairy*, *fermented meat*, or *non-fermented plant*.

A binary presence/absence matrix was first built by pivoting `sample_id` against canonical species. For analyses requiring quantitative comparison to tumour microbiomes, species-level presence/absence was converted to relative abundance and CLR-transformed as described above. The intersection of species between the canonicalised TCGA and cFMD tables defined the shared feature space used for cross-domain embedding and similarity analysis.

4.5 Tumour vs. Normal Comparisons

4.5.1 Prevalence-Based Differential Analysis

Within the TCGA CRC subset, tumour and normal samples were compared using presence/absence information derived from the CLR-transformed abundance matrix. Each species was converted into a binary indicator of detection (above a minimal threshold), allowing the analysis to focus on differences in prevalence rather than absolute CLR magnitude.

For each species, a 2×2 contingency table was constructed, counting the number of tumour and normal samples in which that species was present or absent. Fisher's exact test was applied to each table to determine whether the probability of detecting a given species differed significantly between tumour and normal tissue. P-values were adjusted for multiple comparisons using the Benjamini–Hochberg (BH) false discovery rate procedure.

4.5.2 Presence/Absence Odds Ratios

To quantify effect size, odds ratios (ORs) were computed for each species using the Haldane–Anscombe correction to stabilise estimates in the presence of zero counts. The resulting ORs were $\log_2$-transformed for interpretability, such that positive $\log_2(\mathrm{OR})$ values indicate species more prevalent in tumour tissue and negative values denote enrichment in normal tissue.

For each species, the OR estimates were accompanied by Fisher's exact test p-values, which were adjusted for multiple testing using the Benjamini–Hochberg procedure. Together, the $\log_2(\mathrm{OR})$ distribution and adjusted p-values summarise prevalence-based differences between tumour and normal samples and provide a robust, presence/absence–oriented measure of enrichment.

4.6 Survival-Based Microbial Signatures

4.6.1 Relative Abundance and Delta Metric

For survival analyses, patients were stratified into *survivors* and *deceased* using the curated survival indicator from the TCGA clinical dataset. For each group, species-level relative abundances were computed by normalising counts per sample to percentages. The mean relative abundance per species was then calculated separately for survivors and deceased patients.

A simple effect-size measure was defined as:

$$\Delta_t = \overline{x}_{t,\mathrm{alive}} - \overline{x}_{t,\mathrm{dead}},$$

where $\overline{x}_{t,\mathrm{alive}}$ and $\overline{x}_{t,\mathrm{dead}}$ denote the mean relative abundance of taxon t among survivors and deceased patients, respectively. Positive Δ_t values correspond to taxa enriched in survivors, whereas negative values indicate taxa enriched in deceased patients. Species were ranked by $|\Delta_t|$ to highlight the most strongly shifted taxa, and a summary table of the top taxa was generated.

4.6.2 Shared Food–Tumour Taxa and Survival Enrichment

A key research question was whether species shared between food microbiomes and tumour microbiomes were disproportionately enriched among survivor-associated or deceased-associated taxa. To examine this, the list of shared species obtained from the TCGA–cFMD alignment was intersected with the sets of taxa enriched in survivors and in deceased patients (based on the Δ_t metric and group-wise enrichment tables).

For each shared species, the sign of Δ_t was used to classify it as more abundant in survivors or in deceased patients. Contingency tables comparing "shared vs. non-shared" and "survivor-enriched vs. deceased-enriched" were then analysed using Fisher's exact test, and BH-adjusted p-values were reported. This allowed an assessment of whether food-linked taxa were more frequently associated with favourable or unfavourable survival patterns than expected by chance.

4.7 Cross-Domain Embedding and Similarity

4.7.1 PCA Embedding of TCGA and cFMD Samples

To compare tumour-associated and food-associated microbiomes within a unified framework, the CLR-transformed TCGA and cFMD matrices were restricted to the intersecting set of canonicalised species and concatenated along the sample axis. Principal component analysis (PCA) was performed using `scikit-learn` to obtain a shared low-dimensional embedding:

$$X_{\mathrm{CLR}} \in R^{N\times D} \;\longrightarrow\; Z = X_{\mathrm{CLR}} W,$$

where W contains the leading eigenvectors of the CLR covariance matrix. The first two principal components (PC1 and PC2) were used for visualisation, and the corresponding explained variance ratios were recorded.

Samples from TCGA and cFMD were plotted jointly in the PCA space, with tumour and normal samples distinguished by colour. Food samples were annotated either by fermentation status (F vs. NF) or food type. Additional visualisations presented TCGA samples in the foreground while rendering food samples as a faint background layer, allowing visual assessment of whether particular tumour subgroups lie closer to specific food-associated microbial communities.

4.7.2 Cosine Similarity and Category-Level Clustering

To quantify similarity between tumour, normal, and food categories, mean CLR abundance profiles were computed for each predefined group (e.g. fermented dairy, fermented vegetables, non-fermented meat, tumour tissue, normal tissue). Pairwise cosine similarities were then calculated as:

$$\text{cos_sim}(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|},$$

where u and v denote group-level CLR mean vectors. The similarity matrix was visualised as a heatmap, and in some analyses converted to a distance matrix (1 − similarity) for hierarchical clustering. This provided an interpretable measure of how closely each food category resembles tumour or normal microbial compositions in the shared feature space.

4.7.3 Local Proximity Indices

To characterise finer-grained structure within the PCA embedding, two exploratory proximity metrics were computed. First, a "fermented proximity index" (FPI) was defined as the cosine similarity between each sample's first ten PC scores and the mean PC representation of all fermented food samples. Second, a k-nearest neighbour (kNN) analysis in PCA space quantified, for each sample, the fraction of its nearest neighbours that were food-derived and fermented. These metrics were recorded per sample to identify subsets of tumour samples that cluster locally near fermented food microbiomes within the shared embedding.

4.8 Machine Learning for Survival Prediction

4.8.1 Feature and Target Definition

For supervised prediction of survival status, only primary tumour samples from the TCGA CRC cohort were considered. The feature matrix consisted of batch-adjusted, CLR-transformed species abundances (`tcga_clr_combat1`), and the binary target variable was the curated survival indicator from the TCGA clinical dataset (`alive` vs. `deceased`). Samples with missing survival labels were removed, yielding a dataset containing only cases with complete microbial and clinical information. Because microbial survival prediction is exploratory in nature, the emphasis of the analysis was on identifying taxa contributing to predictive signal rather than on building a clinically deployable classifier.

4.8.2 Train/Test Split and Class Imbalance Handling

Data were randomly partitioned into training (70%) and test (30%) sets using `train_test_split`, stratified by survival status to maintain class proportions. Imbalance between alive and deceased patients was addressed using model-specific weighting schemes:

- For the Random Forest classifier, `class_weight='balanced'` was applied to upweight the minority class.
- For the CatBoost classifier, a `scale_pos_weight` parameter equal to the ratio of deceased to alive cases was supplied to reweight the loss.

These adjustments ensured that patterns associated with the minority class (typically deceased patients) were not suppressed during training.

4.8.3 Random Forest Classifier

A Random Forest classifier [16] with 500 trees (`n_estimators=500`) and `class_weight='balanced'` was trained on the training split. Performance on the held-out test set was quantified using accuracy and the area under the receiver operating characteristic curve (AUC). Feature importances were extracted from the trained model, providing a ranking of taxa according to their contribution to survival discrimination.

The highest-ranking taxa were intersected with the shared TCGA–cFMD species set, and their mean CLR abundances were computed across all food samples and stratified by fermentation status. This enabled identification of "candidate protective or risk taxa" whose tumour-level importance coincides with measurable abundance in fermented or non-fermented foods.

4.8.4 CatBoost Classifier with Early Stopping

To complement the Random Forest model, a gradient-boosted decision tree classifier (CatBoost) was trained on the same feature and target sets. CatBoost is designed to handle heterogeneous feature distributions and class imbalance efficiently [17]. The model was configured with up to 1 000 boosting iterations and applied early stopping after 50 rounds without improvement in validation loss. For this purpose, the training split was divided into an internal training subset and a validation subset (80/20).

Model performance was assessed on the external test set using accuracy, balanced accuracy, recall, F1-score for the alive class, and AUC. Feature importances from the CatBoost model were extracted and cross-referenced with the shared TCGA–cFMD species set, providing an independent set of survival-associated taxa with quantifiable representation in food microbiomes.

4.8.5 Model Diagnostics and Visualisation

For both classifiers, confusion matrices for the test set were visualised using heatmaps to illustrate class-specific prediction patterns. Receiver operating characteristic (ROC) curves for both training and test sets were compared to assess overfitting; substantial divergence between training and test AUC values was interpreted as reduced generalisation ability. Classification reports summarised precision, recall, and F1-scores for both classes. Together, these diagnostics contextualised the stability of the learned decision boundaries and the reliability of taxa identified as predictive features.

4.9 Software and Reproducibility

All analyses were conducted in Python 3.11 using open-source libraries (`pandas`, `numpy`, `scipy`, `scikit-learn`, `matplotlib`, `seaborn`, and CatBoost). Random seeds were fixed for model training and train/test splitting to ensure reproducibility of the results. Intermediate and final result tables (e.g. shared species lists, odds ratio tables, survival-enrichment summaries, and machine learning feature importance tables) were exported as CSV files and serve as the basis for the figures and interpretations reported in the Results and Discussion chapters.

To facilitate full reproducibility of these findings, the complete codebase—including environment specifications, data download scripts, and step-by-step replication instructions—is available in the project repository [18]

5 Results and Findings

5.1 Survival-Associated Tumour Microbiome Signatures (RQ1)

The primary research question examined whether survival in colorectal cancer (CRC) is associated with enrichment of diet-linked microbial taxa in the tumour microbiome. After merging TCGA microbial abundance profiles (S4) with curated clinical metadata, survival information was available for 138 primary tumour samples (25 alive, 113 deceased). All subsequent analyses were performed on this subset.

For each microbial species t, mean relative abundances were computed within survivor and deceased groups. A difference metric,

$$\Delta_t = \overline{x}_{t,\text{alive}} - \overline{x}_{t,\text{dead}},$$

was used to quantify group divergence, with positive values indicating survivor-enrichment and negative values indicating enrichment in deceased patients.

5.1.1 Top Survivor-Enriched Species

Table 5.1 summarises the ten taxa with the largest positive Δ_t. The strongest shifts were observed for *Cutibacterium acnes* ($\Delta_t = 1.023$) and *Escherichia coli* ($\Delta_t = 0.759$). The list notably includes taxa commonly associated with skin or environmental sources (e.g., *C. acnes*, *P. putida*), alongside gut-associated genera such as *E. coli* and *Klebsiella pneumoniae*. Probiotic-associated species (*Lactococcus lactis*, *Lactobacillus gasseri*) were present but exhibited minimal effect sizes ($\Delta_t < 0.02$). Several genera in this list are frequently reported as reagent contaminants in low-biomass sequencing datasets [19], [20].

Table 5.1: Top 10 species enriched in survivors, ranked by Δ_t.

Species (canonical)	Δ_t
Cutibacterium acnes	1.023
Escherichia coli	0.759
Klebsiella pneumoniae	0.635
Pseudomonas putida	0.381
Klebsiella michiganensis	0.234
Stenotrophomonas maltophilia	0.095
Enterococcus faecalis	0.075
Xanthomonas campestris	0.038
Lactococcus lactis	0.012
Lactobacillus gasseri	0.004

5.1.2 Top Deceased-Enriched Species

Table 5.2 presents the ten taxa with the most negative Δ_t, indicating higher relative abundance in deceased patients. The top-ranking species included *Mammaliicoccus lentus* ($\Delta_t = -0.206$), *Pantoea agglomerans*, and *Aeromonas caviae*. Several taxa typically associated with environmental or oral niches appeared in this list, as well as taxa conventionally regarded as beneficial commensals, such as *Bifidobacterium longum* and *Lacticaseibacillus paracasei*. Multiple genera in this group have also been identified as contamination-prone in cancer microbiome studies [19], [20].

Table 5.2: Top 10 species enriched in deceased patients, ranked by Δ_t.

Species (canonical)	Δ_t
Mammaliicoccus lentus	-0.206
Pantoea agglomerans	-0.123
Aeromonas caviae	-0.119
Bifidobacterium longum	-0.082
Lacticaseibacillus paracasei	-0.080
Streptococcus mutans	-0.070
Moraxella osloensis	-0.067
Enterobacter asburiae	-0.050
Lactococcus garvieae	-0.045
Curtobacterium citreum	-0.043

5.1.3 UMAP Projection of Shared Food–Tumour Species

To assess whether survivor- and deceased-enriched taxa cluster in microbial feature space, a UMAP embedding was constructed using CLR-transformed, ComBat-adjusted species profiles restricted to canonical species shared between TCGA and cFMD. After prevalence filtering, 69 shared species remained.

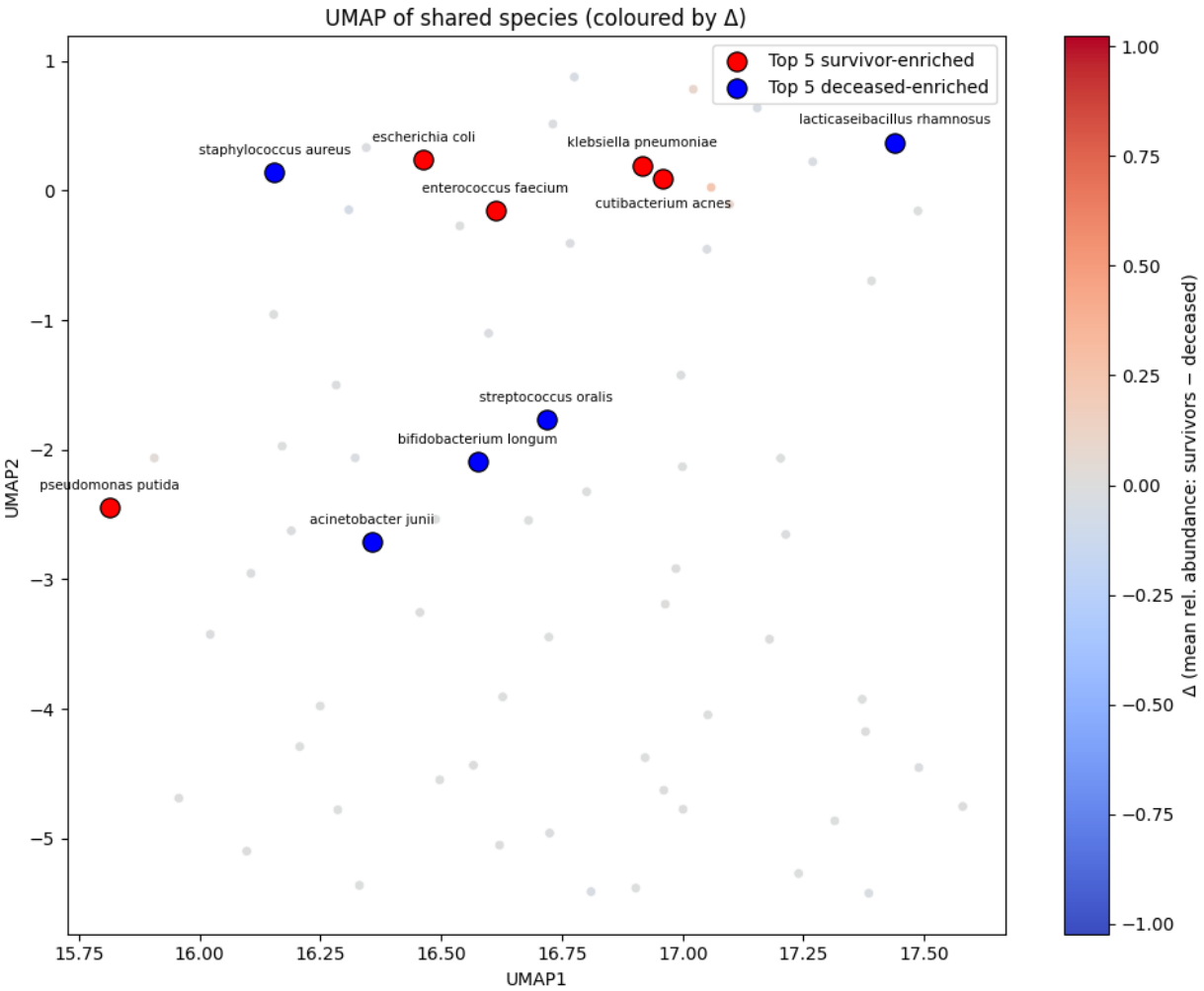

Figure 5.1: UMAP projection of 69 shared TCGA–cFMD species. Each point represents a species; colour denotes Δ_t. Highlighted points indicate the five most survivor-enriched and five most deceased-enriched taxa. No discretely separated clusters are visible.

The UMAP embedding showed no discernible clustering by survival status. Species with positive and negative Δ_t values were interspersed, and the most enriched taxa were distributed broadly across the projection, indicating the absence of shared structural patterns among survival-associated species.

5.1.4 Shared Food–Tumour Taxa and Survival Bias

To determine whether food-associated species exhibited disproportionate survivor- or deceased-enrichment, the analysis was extended to all 313 canonical shared species (prior to prevalence filtering). Species were classified by the sign of Δ_t, and the resulting distribution was compared against all non-shared taxa using Fisher's exact test.

The contingency table for this comparison was:

$$\begin{bmatrix} \text{shared_alive} = 76 & \text{shared_dead} = 195 \\ \text{nonshared_alive} = 1531 & \text{nonshared_dead} = 5516 \end{bmatrix}.$$

Fisher's exact test yielded a nominally significant adjusted p-value (0.0165), driven by a modest odds ratio of 1.404 (Table 5.3). Despite this statistical deviation, neither the magnitude of the effect nor the spatial distribution observed in the UMAP embedding supports a coherent biological enrichment pattern among shared taxa.

Table 5.3: Fisher's exact test comparing survival-enrichment among shared vs. non-shared taxa.

Metric	Value
Odds ratio (shared vs. non-shared)	1.404
FDR-adjusted p-value	0.0165

Overall, the analyses detected no robust or biologically coherent evidence that diet-linked microbial taxa are enriched in survivors of CRC within this dataset.

5.2 Taxonomic Overlap with Food Microbiomes (RQ2)

To address the second research question—whether survival-associated microbial taxa show compositional overlap with specific food categories—the top survivor- and deceased-enriched species identified in RQ1 were cross-referenced with the curated Food Metagenomic Data (cFMD) resource. After canonicalising species names and merging tumour-associated and food-associated taxonomic profiles, 1 518 food samples containing at least one of the RQ1 taxa were identified.

5.2.1 Survivor-Associated Taxa in Dairy Samples

A pronounced pattern emerged for survivor-enriched species, which showed extensive overlap with samples classified as *dairy* within the cFMD metadata. The strongest signal was driven by *Lactococcus lactis*, a lactic-acid bacterium widely used in dairy fermentations and detected in 751 dairy-associated samples (Figure 5.2). This species was also enriched in CRC survivors ($\Delta_t > 0$), producing the clearest taxonomic link between the tumour-associated and food-associated profiles.

Other survivor-enriched species were similarly represented in dairy matrices, including *Escherichia coli* ($n = 33$), *Enterococcus faecalis* ($n = 24$), and *Klebsiella pneumoniae* ($n = 9$), with occurrences in both fermented (F) and non-fermented (NF) dairy products. While these taxa are often interpreted as hygiene indicators or adventitious contaminants in food-processing contexts, their repeated detection in the same food category as *L. lactis* indicates substantial taxonomic overlap between the survivor-associated signature and dairy-associated food microbiomes.

The dominance of *L. lactis* partly reflects the composition of the cFMD database, which contains extensive datasets from dairy-fermentation studies. Nevertheless, the selective enrichment of survivor-associated taxa in the dairy category represents a clear and reproducible pattern.

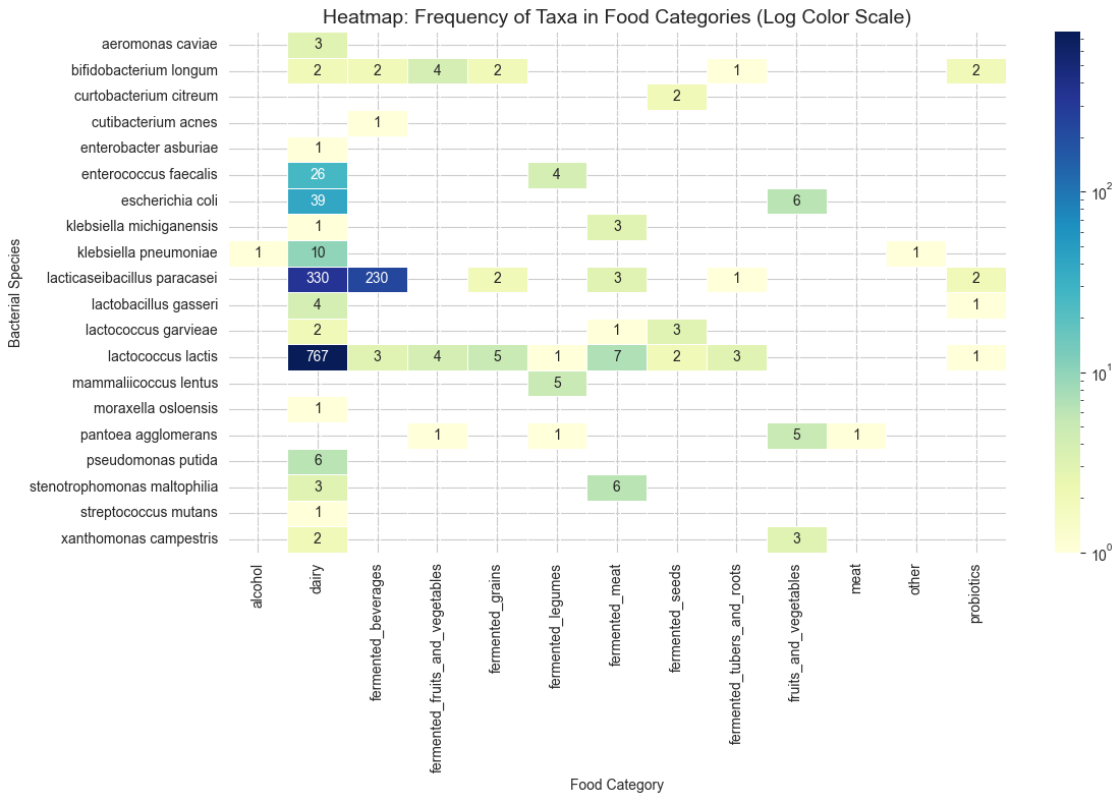

Figure 5.2: Frequency of survivor-enriched taxa detected across food categories in the cFMD database (log scale). *Lactococcus lactis* is overwhelmingly associated with dairy samples ($n = 751$), while other survivor-associated taxa appear at substantially lower frequencies.

Table 5.4 summarises the most frequent food sources for survival-associated taxa, highlighting the strong skew towards dairy products and, to a lesser extent, fermented meats.

Table 5.4: Most frequent food categories containing survivor-associated taxa. Categories and fermentation status are taken directly from cFMD metadata.

Species	Association	Category	Fermented	Count
Lactococcus lactis	Survivor	Dairy	F	751
Escherichia coli	Survivor	Dairy	F	33
Enterococcus faecalis	Survivor	Dairy	F	24
Lactococcus lactis	Survivor	Dairy	NF	16
Klebsiella pneumoniae	Survivor	Dairy	F	9
Lactococcus lactis	Survivor	Meat	F	7
Escherichia coli	Survivor	Dairy	NF	6
Lactobacillus gasseri	Survivor	Dairy	F	4

5.2.2 Scarcity of Food Matches for Deceased-Enriched Taxa

In contrast to the clear overlap observed for survivor-associated taxa, species enriched in deceased patients—such as *Mammaliicoccus lentus*, *Pantoea agglomerans*, and *Aeromonas caviae*—were detected at extremely low frequencies across the cFMD dataset. Their sparse and inconsistent presence across food samples indicates that no coherent dietary reservoir exists for the deceased-associated microbial signature.

Overall, while the survivor-associated taxa show substantial taxonomic overlap with dairy-associated food microbiomes, particularly those involved in lactic-acid fermentation, the taxa enriched in deceased patients do not exhibit a comparable pattern within the cFMD resource.

5.3 Predictive Value of the Tumour Microbiome (RQ3)

The third research question assessed whether tumour-associated microbial profiles contain predictive information regarding patient survival. Machine-learning models were trained on CLR-transformed, ComBat-adjusted species-abundance features from 138 tumour samples with available survival labels (25 alive, 113 deceased). Because of the pronounced class imbalance (approximately 1:4.5), two supervised classifiers—Random Forest and CatBoost—were evaluated on a held-out test set.

5.3.1 Random Forest Classification

The Random Forest classifier achieved an apparent test accuracy of 76%; however, this value was driven almost entirely by the dominance of the majority (deceased) class. Discriminative performance was poor, with a test AUC of 0.574, only slightly above random chance. The model predicted virtually all samples as deceased, resulting in a recall of 0.00 for the survivor class. In contrast, the training AUC reached 1.00, indicating substantial overfitting due to class imbalance and the high dimensionality of the feature space relative to sample size.

Figure 5.3 shows the confusion matrix for the Random Forest model, highlighting the complete failure to correctly classify any survivors.

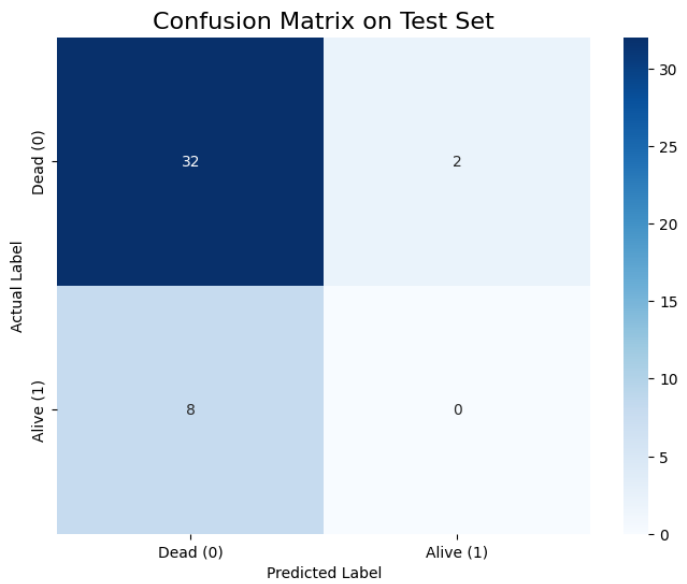

Figure 5.3: Confusion matrix for the Random Forest classifier. All survivor samples were misclassified as deceased.

The ROC curve in Figure 5.4 further illustrates the limited discriminative ability of the model: the test curve lies close to the diagonal corresponding to random guessing, despite perfect separation on the training data.

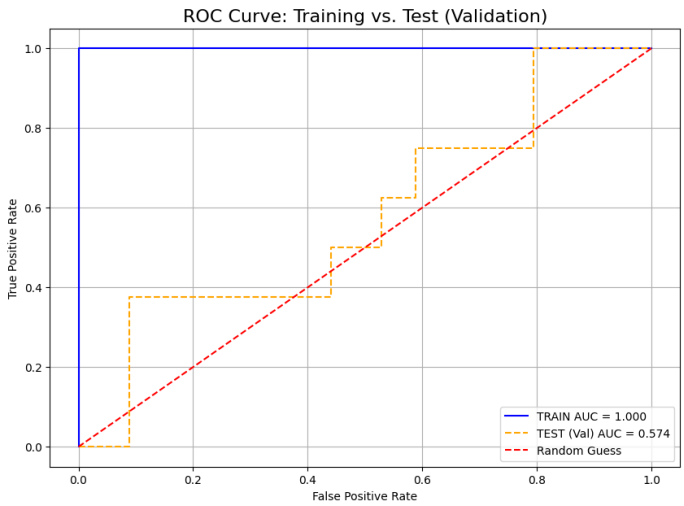

Figure 5.4: ROC curve for the Random Forest classifier. The training curve indicates perfect separation, whereas the test curve remains close to random performance, consistent with severe overfitting.

These results really show how poor this model has performed and it's clear that it cannot discriminate between deseased and alive based on the given microbial composition.

5.3.2 CatBoost Classification

To more directly address class imbalance, a CatBoost classifier with increased weights for the survivor class was trained on the same feature set. This model achieved a balanced accuracy of 52.21%, representing a modest improvement over the Random Forest, but overall discriminative performance remained weak (test AUC = 0.557). The recall for survivors increased to 0.25, yet the majority of survivor samples were still misclassified. As with the Random Forest, CatBoost exhibited near-perfect training performance, again indicating overfitting and poor generalisation.

The confusion matrix in Figure 5.5 shows that only two of the survivors were correctly predicted as alive on the test set.

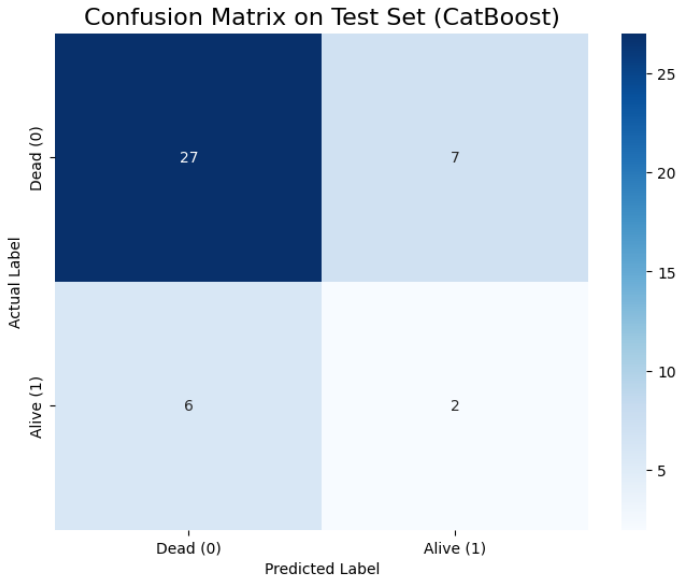

Figure 5.5: Confusion matrix for the CatBoost classifier. Only two survivor samples were correctly identified; most were misclassified as deceased.

The ROC curve in Figure 5.6 confirms the limited discriminative ability of CatBoost: the test curve remains close to the diagonal despite strong separation on the training set, again reflecting overfitting.

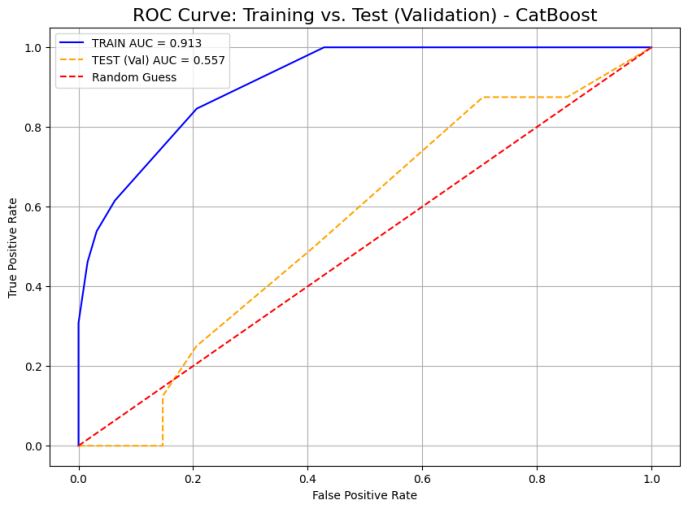

Figure 5.6: ROC curve for the CatBoost classifier. The model overfits the training data but achieves only marginally better-than-random discrimination on the test set.

5.3.3 Association Between Microbiome Structure and Tumour Stage

Because survival is partly determined by tumour progression, an auxiliary analysis tested whether tumour microbiome structure was associated with pathological tumour stage. An ordinary least squares regression using the first five principal components of the CLR-transformed species matrix explained only 4.1% of the variance ($R^2 = 0.041$, $p = 0.296$). This indicates that tumour-resident microbial composition did not show a meaningful linear association with disease severity in this cohort.

5.3.4 Summary of Predictive Analysis

Across both non-linear classifiers and linear regression, the tumour microbiome showed no meaningful predictive capacity for survival outcomes. All models yielded low AUC values (< 0.60), poor recall for the survivor class, and clear evidence of overfitting. These findings mirror the absence of structured differences in the UMAP embedding (RQ1) and indicate that, within this dataset, species-level tumour microbiome profiles do not provide a reliable basis for predicting survival in colorectal cancer.

6 Discussion

The objective of this study was to investigate whether tumour-associated microbial profiles in colorectal cancer (CRC) show evidence of dietary influence and whether such patterns relate to patient survival. By integrating species-level microbial features from TCGA with food-associated microbiomes from the curatedFoodMetagenomicData (cFMD) resource, three research questions were examined: (1) whether survival is associated with enrichment of diet-linked microbial taxa (RQ1), (2) whether survival-associated taxa overlap with particular food categories (RQ2), and (3) whether tumour microbial profiles contain predictive information for survival outcomes (RQ3). Across all analyses, no robust or biologically coherent evidence was found to support diet-linked survival signatures in this dataset.

6.1 Summary of Principal Findings

Three major results emerged:

1. Differential abundance analysis (RQ1) identified apparent shifts between survivors and deceased patients; however, the taxa driving these differences were dominated by contamination-prone genera and exhibited biologically implausible effect directions.
2. Survivor-associated taxa overlapped strongly with dairy-associated microbes in the cFMD dataset (RQ2), particularly fermented dairy species, but this pattern reflected ecological ubiquity rather than dietary transfer into tumour tissue.
3. Machine-learning models (RQ3) showed no predictive power for survival, consistent with the weak and noise-driven microbial signals identified in RQ1.

These findings indicate that TCGA tumour microbiome profiles do not contain reliable information about dietary origins or survival outcomes.

6.2 Interpretation of Survival-Associated Microbial Signals (RQ1)

The differential abundance analysis highlighted several taxa enriched in survivors or deceased patients. However, multiple lines of evidence indicate that these patterns do not reflect biological tumour-associated microbiota.

6.2.1 Contamination-prone and environmentally ubiquitous taxa

Several of the top survivor-enriched taxa belong to genera strongly associated with reagent or environmental contamination:

- ***Cutibacterium acnes*** — a dominant skin commensal and one of the most frequently reported contaminants in sequencing workflows [19]. Its presence in tissue WGS data is widely interpreted as a technical artifact [20].
- ***Pseudomonas putida***, ***Stenotrophomonas maltophilia***, ***Xanthomonas campestris*** — environmental bacteria often introduced during sample handling or reagent preparation [19].

The apparent enrichment of these taxa among survivors therefore most likely reflects technical variability rather than biological differences.

6.2.2 Taxa with known pathogenic or tumour-promoting potential

Several taxa enriched in survivors contradict established CRC-related microbial mechanisms:

- ***Escherichia coli*** — strains encoding the colibactin gene cluster induce DNA double-strand breaks and promote tumorigenesis [21]. Survivor enrichment is therefore biologically implausible.
- ***Enterococcus faecalis*** — produces extracellular superoxide and hydrogen peroxide that damage host DNA [22]. Again, survivor enrichment contradicts mechanistic expectations.
- ***Klebsiella pneumoniae*** and ***Klebsiella michiganensis*** — opportunistic pathogens commonly detected as contaminants in low-biomass sequencing [19].

These inverted effect directions further support the interpretation that the observed patterns arise from noise rather than functional tumour ecology.

6.2.3 Beneficial taxa with negligible effect sizes

Two taxa are notable because external literature reports potential tumour-suppressive or immunomodulatory effects:

- ***Lactobacillus gasseri*** — recent work has demonstrated anti-inflammatory and anti-proliferative effects in CRC models and improved survival in murine experiments [23].
- ***Lactococcus lactis*** — used as a probiotic delivery system and known to modulate mucosal immunity [24].

However, in this dataset, their estimated effect sizes were extremely small ($\Delta_t < 0.02$). Thus, even though external work suggests potential benefits, the TCGA-derived microbial profiles do *not* provide statistically or ecologically coherent evidence supporting their involvement in CRC survival.

6.2.4 Lack of ecological signal

The UMAP embedding of shared species (Fig. 5.1) showed no structural separation between survivor- and deceased-enriched taxa. Instead, taxa were diffusely intermixed, indicating absence of coherent ecological states. If biologically meaningful microbial differences existed, one would expect separable clusters or spatial gradients.

Additionally, statistical power was limited due to the small survivor group (25 samples), reducing stability of effect estimates and increasing the likelihood of spurious associations.

6.3 Diet–Tumour Microbial Overlap with Food Microbiomes (RQ2)

Cross-referencing RQ1 taxa with cFMD species annotations revealed 1 518 food samples containing at least one of the survivor- or deceased-associated species.

6.3.1 Survivor-associated taxa overlap with dairy-fermentation species

A pronounced overlap was observed between survivor-enriched taxa and dairy-associated species, especially fermented dairy. *Lactococcus lactis*—a canonical cheese and yoghurt starter culture—appeared in 751 dairy samples and was enriched in survivors. This overlap is ecologically plausible: dairy-rich studies constitute a large portion of cFMD, and *L. lactis* is ubiquitous in such environments.

However, this does not imply dietary transfer to tumour tissue. Several of the overlapping taxa (*E. coli*, *K. pneumoniae*, *E. faecalis*) are well-known hygiene indicators or environmental contaminants, whose presence reflects shared ecological niches rather than diet-mediated

tumour colonisation.

6.3.2 Absence of dietary coherence among deceased-associated taxa

Species enriched in deceased patients showed extremely sparse and inconsistent matches across food samples, most originating from environmental or host-associated reservoirs rather than dietary sources. No coherent food category emerged for this group.

6.3.3 Interpreting the Fisher's exact test

While Fisher's exact test yielded nominal statistical significance (OR = 1.404, FDR-adjusted $p = 0.0165$), this effect lacked biological or ecological coherence. The signal is best interpreted as arising from large denominator differences and dataset size rather than meaningful diet–tumour relationships.

6.4 Predictive Value of the Tumour Microbiome (RQ3)

Machine-learning classifiers (Random Forest, CatBoost) trained on CLR-transformed species abundances showed:

- very low discriminative power (AUC < 0.60),
- severe overfitting (training AUC 1.00),
- and poor recall for the survivor class.

These results mirror the weak, noise-driven microbial signals described in RQ1 and align with independent evaluations showing that TCGA tumour microbiomes carry little predictive information [5], [25].

A supplementary OLS regression found no association between microbial principal components and tumour stage, indicating that species-level signatures do not reflect tumour progression.

6.5 Limitations

Several factors constrain the interpretation of the results obtained in this report:

- **Limited Sample Size and Statistical Power:** After quality filtering and dataset merging, only 25 patients were classified as survivors. This small minority group reduces effect-size stability, limits the ability to detect subtle microbial differences, and contributes to model overfitting in supervised learning.
- **Binary Survival Labels:** Survival was analysed as a binary outcome (alive vs. deceased), despite the availability of clinical follow-up durations.The absence of time-to-event modelling restricts interpretability and prevents evaluation of whether microbial signals relate to survival time rather than simple group membership.
- **Indirect Dietary Inference:** cFMD provides microbial compositions of food items, not dietary intake. Overlaps between tumour- and food-associated taxa therefore reflect ecological similarity or database composition rather than confirmed patient consumption. This prevents causal or behavioural interpretation of diet–tumour relationships.
- **Species-Level Resolution:** All analyses were conducted at the species level, which cannot distinguish between strains with contrasting functional profiles. This is particularly important for taxa such as *Escherichia coli* and *Enterococcus faecalis*, where pathogenicity, genotoxicity, or probiotic potential depend on strain-specific features. Species-level

associations should therefore be interpreted with caution.

- **Potential Contamination in Low-Biomass Samples:** TCGA tumour microbiome data lack negative controls, spike-ins, and laboratory metadata. Several taxa identified as survival-associated have previously been described as reagent contaminants, making it difficult to distinguish true tumour colonisers from technical noise.

6.6 Implications and Future Directions

These findings highlight several methodological priorities for future research on tumour-associated microbiomes and potential dietary influences:

- **Paired Sample Designs:** Collecting matched stool, tumour, and blood samples would help distinguish resident microbial signals from transient or contaminant taxa and clarify microbial origins.
- **Integration of Dietary Metadata:** Incorporating validated dietary intake records alongside food-metagenome databases would allow direct testing of diet–microbiome–tumour pathways rather than relying on ecological overlap.
- **Quality Filtering of Food-Microbiome Data:** When mapping taxa between tumour samples and cFMD, future analyses should incorporate MAG contamination scores, completeness thresholds, and culture-based validation to improve source attribution.
- **Longitudinal and Mechanistic Studies:** Establishing causal relationships will require controlled dietary interventions, tracking of microbial strain transfer, and functional assays investigating microbial effects on tumour biology.

7 Conclusion

This study investigated potential links between diet-associated microbial taxa, the tumour microbiome, and survival outcomes in colorectal cancer (CRC) by integrating tumour-derived microbial profiles from TCGA with food-associated metagenomes from the curatedFoodMetagenomicData (cFMD) resource. Three research questions were addressed: whether survival is associated with diet-linked microbial enrichment (RQ1), whether survival-associated taxa overlap with specific food categories (RQ2), and whether tumour microbial compositions can predict patient survival (RQ3).

Across all analyses, no robust evidence was found to support a diet–microbiome–survival relationship in this dataset. The differential abundance contrasts between survivors and deceased patients were dominated by contamination-prone genera, biologically implausible effect directions, and noise typical of low-biomass tumour sequencing. UMAP embeddings and Fisher's exact testing further showed that these patterns lacked ecological coherence and did not reflect meaningful microbial structuring.

Cross-referencing survival-associated taxa with the cFMD database revealed substantial overlap between survivor-enriched species and dairy-associated microbes—particularly *Lactococcus lactis*. However, this pattern reflected ecological ubiquity within fermented dairy products rather than credible evidence of dietary transfer into tumour tissue. Deceased-associated taxa did not correspond to any food category, reinforcing the absence of dietary relevance in tumour microbial profiles.

Predictive modelling using Random Forest and CatBoost classifiers demonstrated poor discrimination between survivors and deceased patients, with AUC values below 0.60 and clear signs of overfitting. Tumour microbial composition also showed no association with tumour stage, further indicating that species-level microbial profiles provide limited clinical information within the TCGA cohort.

Taken together, these findings show that tumour-associated microbial data from TCGA do not support hypotheses linking diet-derived taxa to survival outcomes in CRC. Instead, they highlight the methodological challenges of analysing low-biomass tumour microbiomes, the need for stringent contamination-aware sequencing workflows, and the importance of integrating strain-level resolution, functional data, and validated dietary metadata in future research. Robust investigation of diet–tumour–microbiome interactions will require purpose-designed clinical studies rather than retrospective analyses of existing sequencing datasets.

8 Appendix

8.1 Project Repository and Code Availability

The complete source code, including data preprocessing pipelines, statistical analysis scripts, and machine learning models, is available at the following public repository:

https://github.com/acg-team/PA2-Dietary-Linkage-to-Microbiome-and-Their-Influence-on-Colorectal-Cancer

8.2 Declaration of AI Assistance

In accordance with the ZHAW guidelines on the use of artificial intelligence in academic work, I declare that AI-based tools were used throughout the preparation of this report. The following systems were employed between September and December 2025:

- **ChatGPT Pro (GPT-5 / 5.1)** Used for support in data analysis, debugging, code generation, optimisation of Python workflows, interpretation of statistical and machine-learning outputs, structuring of results, and drafting and refining written sections.
- **Google Gemini Advanced (Gemini 2.5 / 3.0)** Used for explaining complex methodological steps, cross-checking interpretations, validating biological background information, and improving clarity in the presentation of analytical reasoning.
- **Perplexity Pro** Used as a research assistant for literature exploration, summarising scientific publications, comparing methodological approaches, and verifying conceptual consistency.

These tools served as *assistive* technologies to enhance efficiency, support coding tasks, provide alternative explanations, and improve the linguistic quality of the manuscript. AI was also used iteratively during planning, analysis interpretation, and report writing.

All AI-generated outputs were critically reviewed, adapted, or rewritten by me. Decisions regarding methodology, data processing, statistical interpretation, and scientific conclusions are in my responsibility.

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